Agreement Among Explainability Methods: A Cross-Model, Cross-Domain Study

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Feature-attribution methods such as SHAP, LIME, and permutation importance are routinely used to justify machine learning decisions, on the implicit assumption that they largely agree on which features matter. We test that assumption directly: across four tabular datasets covering three domains (two educational, one financial, one agricultural), three model families (random forest, gradient boosting, logistic regression), and ten cross-validation folds, we compute 360 feature-importance rankings and compare them pairwise using Kendall's τ along three axes: between folds of the same method, between methods applied to the same model, and between models explained by the same method. Agreement between folds of the same method is consistently high (mean τ between 0.54 and 0.89, at or above 0.75 for three of the four datasets), but agreement between different explanation methods applied to the same model is substantially and significantly lower (mean τ between 0.28 and 0.47), with model-to-model agreement for a fixed method falling in between (0.50 to 0.55). Crucially, agreement drops even further (e.g., $\tau = 0.055$ on German Credit) when restricted to the top-10 features actually reported to stakeholders. A two-way ANOVA confirms both the dataset and the comparison-axis effects are highly significant ($p < 10^{-120}$), with a significant interaction. Which explanation method is chosen matters more than which model is explained, and reporting the output of a single attribution method without checking it against another can misrepresent which features a model relies on.

Povzetek: Metode za razlago napovedi modelov strojnega učenja, kot so SHAP, LIME in permutacijska pomembnost, se pogosto uporabljajo ob predpostavki, da se njihove ocene pomembnosti atributov medsebojno ujemajo. V tej empirični študiji na štirih tabelaričnih naborih podatkov, treh družinah modelov in desetih rezinah prečnega preverjanja neposredno merimo skladnost med metodami s pomočjo Kendallovega koeficienta τ. Rezultati kažejo, da je stabilnost posamezne metode visoka, medtem ko je ujemanje med različnimi metodami na istem modelu bistveno nižje, kar poudarja potrebo po hkratni uporabi več metod razložljivosti v praksi.

1 Introduction

Attribution-based explainability methods are increasingly used as a substitute for genuine model transparency: rather than build inherently interpretable models, practitioners train a black box and then ask a post-hoc method (most commonly SHAP, LIME, or permutation importance) to say which features drove a prediction. The implicit premise behind this practice is that these methods largely converge on the same answer, so that the choice of which one to run is a matter of convenience rather than substance. That premise is rarely tested; most applied papers report the output of a single method without checking whether a different, equally legitimate method would have

told a different story.

We test the premise directly, and deliberately across dissimilar domains, so that any inconsistency we find cannot be dismissed as an artifact of one dataset's idiosyncrasies. We ask three questions. First, how stable is a single explanation method across cross-validation folds of the same model (folds share most of their training data with each other under k-fold CV, so this is a lower bound on true independent-sample stability, not a fully independent replication): how much of the ranking is signal versus resampling noise? Second, holding the model fixed, how much do SHAP, LIME, and permutation importance agree with each other? Third, holding the explanation method fixed, how much does the ranking change when the underlying model changes? If explanation methods were interchangeable, the second quantity should be close to the first. We show that it is not.

The contributions of this work are: (1) a controlled, quantitative measurement of explanation-method disagreement across four datasets from unrelated domains, three model families, and ten cross-validation folds, decomposed into fold-stability, method-disagreement, and model-disagreement axes; (2) a factorial ANOVA confirming that the dataset and comparison-axis effects on agreement are both highly significant, with a significant interaction, so the pattern generalizes across domains rather than reflecting one dataset's idiosyncrasies; (3) a practical demonstration, via top-10 feature-ranking comparisons, that this statistical disagreement translates into visibly different feature lists depending on which attribution method produced them.

2 Related Work

SHAP [1] and LIME [2] are the two dominant post-hoc attribution methods in applied machine learning, each grounded in a different theoretical framework (Shapley-value game-theoretic allocation for SHAP, local linear surrogate models for LIME) that gives no a priori guarantee the two will agree in practice. Permutation importance, the third method we use, was formalized as model reliance by Fisher, Rudin and Dominici [3], who additionally showed that importance rankings can vary substantially across different well-performing models fit to the same data (their "Rashomon set" framing), a result that anticipates, from a different angle, our between-model comparison axis. The theoretical plurality of these frameworks is not accidental: Lipton [4] argues that "interpretability" is a term without consensus definition, covering at minimum three distinct properties (simulability, decomposability, and algorithmic transparency) that are neither mutually implied nor uniformly relevant to every use case; from this vantage point, SHAP, LIME, and permutation importance each measure a genuinely different thing, and disagreement between them is not a bug to be corrected but a structural fact about the methods.

Most directly relevant, Krishna et al. [5] name and formalize what they call the disagreement problem in explainable machine learning, interviewing practicing data scientists and reporting that widely adopted explanation methods frequently disagree, with practitioners resolving conflicts through ad hoc heuristics rather than principled criteria. Our study can be read as a controlled, quantitative companion to that practitioner-survey finding: rather than ask analysts how they perceive disagreement, we measure it directly, at scale (360 rankings, four domains, ten-fold cross-validation, bootstrap confidence intervals and a factorial ANOVA), and decompose it into the fold-stability, method-disagreement, and model-disagreement axes that a practitioner-facing study cannot easily separate. Where their contribution is to establish that the problem is real and consequential in practice, ours is to quantify its magnitude precisely and show it replicates across unrelated application domains. An earlier and complementary line of work by Alvarez-Melis and Jaakkola [6] showed that LIME and SHAP are individually brittle to small perturbations of the input, and proposed local sensitivity metrics to quantify this; our repeated-seed control (Results) addresses a different, weaker form of instability, namely sampling noise within a fixed input, and shows it does not account for the between-method disagreement we report. The disagreement we measure, then, is not reducible to per-method noise: it reflects the fact that each method is recovering a structurally different concept of "importance." Doshi-Velez and Kim [7] argue that interpretability methods should be

held to the same evidentiary standards as other modeling outputs; applied to attribution rankings, this implies that a ranking presented without a concordance check is not validated in the same sense that a predictive model cross-validated against held-out data is validated. Our τ-based concordance measurement is one concrete operationalisation of that principle.

Empirical applications of explanation methods in the literature, meanwhile, typically report a single method's output without checking it against an alternative. In the recent literature of this journal, post-hoc attribution and explainable representations are actively applied across critical domains: Li [8] combines XGBoost and SHAP to extract key behavioral factors driving student academic outcomes in large-scale educational records, directly reflecting the pedagogical context of our educational benchmarks (OULAD and Dropout); similarly, Yao and Zhu [9] utilize Random Forest with SHAP to explain dynamic cost drivers in institutional resource allocation, mirroring the tabular risk-modeling setup of financial benchmarks like German Credit; and Vlahek [10] develops iterative algorithms for explainable feature learning to balance predictive accuracy and interpretability. While these contributions demonstrate the operational value of XAI, they standardly rely on a single attribution paradigm without evaluating whether alternative explanation frameworks (such as LIME or permutation importance) would report concordant feature hierarchies. Our study directly complements this literature by providing a systematic, quantitative assessment of inter-method concordance across these exact model families and domain structures. A systematic reference for the methods themselves is provided by Molnar [11], whose treatment of SHAP, LIME, and permutation importance in a unified framework is the source for the aggregation procedure we use for LIME's local-to-global ranking (Chapter 10 of that text).

The reliability concerns motivating this study are not hypothetical. Slack et al. [12] demonstrate that an adversary can construct a classifier whose biased predictions are invisible to LIME and SHAP by exploiting the perturbation-based local sampling they rely on; if methods can be systematically misled in adversarial conditions, ordinary practitioners operating in good faith face the subtler but structurally analogous risk that two methods evaluated on the same instance will report different rankings for non-adversarial reasons. Kindermans et al. [13] demonstrate formally that widely used saliency methods can produce completely different attributions under trivial input transformations (e.g., a constant shift), establishing that the disagreement between explanation methods is not merely an empirical observation but a theoretical property of the methods themselves; our Kendall's τ measurements across 360 rankings provide a controlled, quantitative estimate of the magnitude of this theoretical prediction on tabular data. Nauta et al. [14] provide a systematic review of quantitative evaluation methods for explainable AI, documenting that the majority of studies report explanations without validating concordance between methods, and proposing a taxonomy of evaluation properties (faithfulness, robustness, complexity) that our τ-based concordance measurement directly instantiates for the property they call "agreement." Visani, Bagli and Chesani [15] quantify this directly by introducing statistical stability indices for LIME, showing that the variance across repeated calls with different random seeds is high enough for individual feature rankings to reverse — a finding we evaluate in the Results section through a repeated-seed robustness check across five distinct sampling seeds, demonstrating that same-method sampling variance ($\tau \geq 0.94$) is negligible compared to the observed between-method divergence. Rudin [16] argues, from a theoretical standpoint, that no post-hoc approximation of a black-box model can be fully faithful, and therefore any explanation is, by construction, a partially incorrect summary of the model; our empirical observation that LIME, SHAP, and permutation importance disagree at the instance level is a direct, quantitative consequence of that theoretical argument. Speith [17] provides a recent systematic taxonomy of 27 XAI classification schemes, confirming that SHAP, LIME, and permutation importance are typically grouped as "post-hoc model-agnostic local" methods and noting that cross-method empirical comparisons of their outputs remain rare in the literature — a gap this paper fills.

3 Methodology

3.1 Datasets

We use four public tabular datasets (characteristics in Table 1), chosen for domain diversity so that a finding replicated across all four cannot be attributed to a property of a single domain: the Open University Learning Analytics Dataset (OULAD, education, final result classification, four original classes: Pass/Fail/Withdrawn/Distinction), the UCI "Predict Students' Dropout and Academic Success" dataset (education, a different institutional context, three classes: Dropout/Enrolled/Graduate), the Statlog German Credit dataset (finance, binary), and the Rice (Cammeo and Osmancik) dataset (agriculture, binary classification by grain morphology). All four datasets retain their original class structure: we deliberately did not collapse the multi-class targets to binary, so that the multi-class attribution-aggregation procedure described below is itself exercised and reported, rather than defined away.

3.2 Models and attribution methods

For each dataset we train three models, random forest [18], gradient boosting (XGBoost) [19], and logistic regression, under 10-fold stratified cross-validation, using identical preprocessing (median/most-frequent imputation, standard scaling for numeric features, one-hot encoding for categorical features) across all three so that differences in attribution cannot be attributed to preprocessing differences. Random forest depth was capped at 12 splits; uncapped trees made SHAP computation intractable on the larger dataset (OULAD) without any compensating gain in predictive accuracy. For each (dataset, model, fold), we compute three feature-attribution rankings: SHAP (TreeExplainer for tree models, LinearExplainer for logistic regression), LIME (a random sample of test instances explained locally, weights aggregated by class-predicted-label to build one global ranking per fold), and permutation importance [3] (on a bounded random subsample of the test fold, for tractability, following the same subsampling principle the SHAP computation already required). For multiclass datasets, SHAP and permutation attributions are aggregated across classes by averaging the absolute per-class attribution before ranking; for LIME, each instance's explanation is taken with respect to its own predicted class before aggregation. This yields 3 models $\times$ 3 methods $\times$ 10 folds $\times$ 4 datasets = 360 rankings.

3.3 Predictive performance

Because disagreement between explanation methods is only interesting relative to how much the explained model itself has actually learned, we report each model's held-out predictive performance under the same 10-fold cross-validation used for the attribution comparisons (Table 2). Performance is well above chance ($1/n_{\text{classes}}$) for all four datasets, but far from uniform: models on Rice are strong (accuracy $>$ 0.91), German Credit and Dropout are moderate (accuracy 0.75–0.78), and OULAD is comparatively weak (accuracy 0.44–0.45 on four classes, chance = 0.25) despite being the largest dataset by sample size. Model choice barely affects performance within a dataset (accuracy spans 1.0–1.4 percentage points across the three models on every dataset), meaning the between-model τ axis in Table 3 compares explanation rankings across models of essentially equivalent predictive quality, not a strong model against a weak one. OULAD's comparatively weak absolute performance is worth keeping in mind when interpreting that dataset's attribution rankings: a model that predicts only modestly better than chance has less genuine signal for any explanation method to recover, so some of OULAD's between-method disagreement could reflect the model's own limited signal rather than a property of the explanation methods themselves.

3.4 Agreement metric

We measure pairwise ranking agreement with Kendall's τ and organize comparisons along three axes: *between folds* (same model, same method, different fold; a measure of the method's own stability), *between methods* (same model, same fold, different method; the central quantity of interest), and *between models* (same method, same fold, different model; whether an explanation method's conclusions generalize across model families). For

Table 1: Characteristics of the four datasets before pipeline preprocessing (one-hot encoding of categorical predictors increases the feature count actually seen by each model).

Dataset	Domain	Task	n	Predictors	Class balance
OULAD	Education	4-class	32,593	49	Pass 37.9% / Withdrawn 31.2% / Fail 21.6% / Distinction 9.3%
Dropout	Education	3-class	4,424	36	Graduate 49.9% / Dropout 32.1% / Enrolled 17.9%
German Credit	Finance	Binary	1,000	61	Good risk 70.0% / Bad risk 30.0%
Rice	Agriculture	Binary	3,810	7	Osmancik 57.2% / Cammeo 42.8%

Table 2: Held-out predictive performance (mean ± SD across 10 folds), same cross-validation split used throughout.

Dataset	Model	Accuracy	F1 (macro)	Log-loss
OULAD	Random forest	0.446 ± 0.006	0.286 ± 0.006	1.212 ± 0.005
	Gradient boosting	0.440 ± 0.006	0.320 ± 0.005	1.223 ± 0.009
	Logistic regression	0.450 ± 0.010	0.307 ± 0.009	1.205 ± 0.006
Dropout	Random forest	0.775 ± 0.016	0.692 ± 0.023	0.572 ± 0.022
	Gradient boosting	0.777 ± 0.017	0.710 ± 0.021	0.616 ± 0.058
	Logistic regression	0.763 ± 0.013	0.673 ± 0.022	0.577 ± 0.024
German Credit	Random forest	0.758 ± 0.033	0.656 ± 0.049	0.492 ± 0.028
	Gradient boosting	0.765 ± 0.024	0.698 ± 0.041	0.620 ± 0.078
	Logistic regression	0.751 ± 0.029	0.680 ± 0.040	0.502 ± 0.043
Rice	Random forest	0.927 ± 0.013	0.925 ± 0.013	0.246 ± 0.058
	Gradient boosting	0.916 ± 0.010	0.914 ± 0.010	0.251 ± 0.035
	Logistic regression	0.929 ± 0.012	0.928 ± 0.012	0.183 ± 0.017

each axis we report the mean and median τ with a 2,000-resample bootstrap 95% confidence interval, and we fit a two-way ANOVA of τ on dataset and comparison axis (with interaction) to test whether the pattern is consistent across datasets. The 2,340 pairwise τ values feeding that ANOVA are not fully independent: the 405 between-fold pairs per dataset, for example, are built from only 10 underlying fold rankings per (model, method), so as a robustness check we repeat the ANOVA on cluster means (one τ per dataset, axis, and model/method-pair combination, collapsing 2,340 values to 276), which reduces pseudo-replication substantially; both the axis and dataset effects remain highly significant under this more conservative specification (Results).

4 Results

The pattern in Table 3 and Figures 1, 2, and 3 is consistent across all four datasets: stability across folds of the *same* method is high (mean τ from 0.54 to 0.89), agreement *between* methods applied to the same model is roughly half that (mean τ from 0.28 to 0.47), and agreement between models for a fixed method sits in between (0.50 to 0.55). A two-way ANOVA of τ on dataset and comparison axis (Table 4) confirms both main effects are highly significant (dataset: $F = 222.6$, $p < 10^{-126}$, partial $\eta^2 = 0.223$; axis: $F = 527.1$, $p < 10^{-189}$, partial $\eta^2 = 0.312$) with a significant but comparatively small interaction ($F = 18.9$, $p < 10^{-20}$, partial $\eta^2 = 0.047$), meaning the magnitude of the effect depends somewhat on the dataset but its direction and general size do not, and the comparison axis (fold/method/model) accounts for a somewhat larger share of variance in τ than the dataset does. Because the raw τ values are clustered (many pairs share the same underlying fold rankings), we repeat this test on the cluster-mean specification described in Methodology; both main effects remain highly significant (dataset: $F = 22.5$, $p < 10^{-12}$; axis: $F = 122.4$, $p < 10^{-37}$), confirming the pattern is not an

Table 3: Mean Kendall's τ by comparison axis and dataset (bootstrap 95% CI in brackets).

Dataset	Between folds	Between methods	Between models
OULAD	0.75 [0.73, 0.76]	0.46 [0.44, 0.48]	0.55 [0.52, 0.57]
Dropout	0.76 [0.75, 0.78]	0.41 [0.38, 0.45]	0.55 [0.53, 0.57]
German Credit	0.54 [0.51, 0.57]	0.28 [0.26, 0.31]	0.50 [0.47, 0.53]
Rice	0.89 [0.88, 0.91]	0.47 [0.40, 0.53]	0.55 [0.50, 0.60]

artifact of treating correlated pairwise comparisons as independent observations. The between-method agreement for the Rice dataset exhibits a particularly wide confidence interval and includes negative τ values in Figure 1; this is an expected artifact of its low dimensionality (only 7 features), which makes τ highly volatile and capable of crossing zero even with minor rank perturbations.

The number of features per dataset ranges from 7 (Rice) to 61 (German Credit, post-encoding), and τ on the full ranked list could plausibly overstate agreement for the higher-dimensional datasets, since most ranked positions there are low-importance features whose relative order is close to noise and may agree by chance. To check this, we recomputed between-method τ restricted to the union of each pair's top-10 features (the practically relevant portion of the ranking for a stakeholder reading off the most important predictors), for the between-method axis specifically. The result runs opposite to the noise-inflation concern: restricting to the top-10 *reduces* agreement for the three higher-dimensional datasets rather than increasing it (dropout: 0.414 full-list vs. 0.244 top-10; German Credit: 0.284 vs. 0.055; OULAD: 0.461 vs. 0.189), while Rice, whose full feature list is already close to 10, is essentially unchanged (0.467 vs. 0.467). Agreement among SHAP, LIME, and permutation importance on precisely the features most likely to be reported to a stakeholder is therefore lower, not higher, than the full-list τ values in Table 3 suggest, for three of the four datasets.

Because LIME and permutation importance are both estimated from a random subsample of test instances rather than computed exactly, part of the between-method disagreement we report could in principle be estimator noise rather than genuine disagreement between the underlying methods. We tested this directly with a repeated-seed control: holding the trained model, dataset, and fold fixed, we recomputed each method's ranking five times with different sampling seeds and measured τ between repetitions of the *same* method. If between-method τ were mostly estimator noise, this same-method τ should be comparably low. Instead it is high: mean $\tau = 0.970$ for LIME and 0.915 for permutation importance across the four datasets (only OULAD's permutation-importance repetitions show any real spread, mean $\tau = 0.660$; all other dataset-method combinations exceed 0.93, several reaching 1.0). Estimator noise is therefore a minor contributor to the reported between-method disagreement (mean $\tau \in [0.28, 0.47]$): re-running the same method with a different seed reproduces essentially the same ranking, while switching methods does not.

Put plainly: if two researchers explain the same trained random forest, one with SHAP and one with LIME, their rankings of which features matter will typically show 64%–74% of feature pairs in the same relative order (Kendall's $\tau \approx 0.28$–0.47, corresponding to $(\tau + 1)/2$ concordant pairs), while the same researcher re-running the same method on a different resampling fold of the same data will get a ranking that agrees with their first run substantially more (77%–95% concordant pairs, $\tau \approx 0.54$–0.89, and at or above 0.75 for three of the four datasets). Choice of model family matters comparatively less than choice of explanation method: switching from a random forest to logistic regression while keeping the explanation method fixed produces disagreement of about the same magnitude ($\tau \approx 0.50$–0.55) as switching explanation methods while keeping the model fixed ($\tau \approx 0.28$–0.47); both axes fall well below the within-method stability ceiling.

Figure 4 makes the disagreement concrete rather than only statistical: for the random forest model, several features that rank in SHAP's top

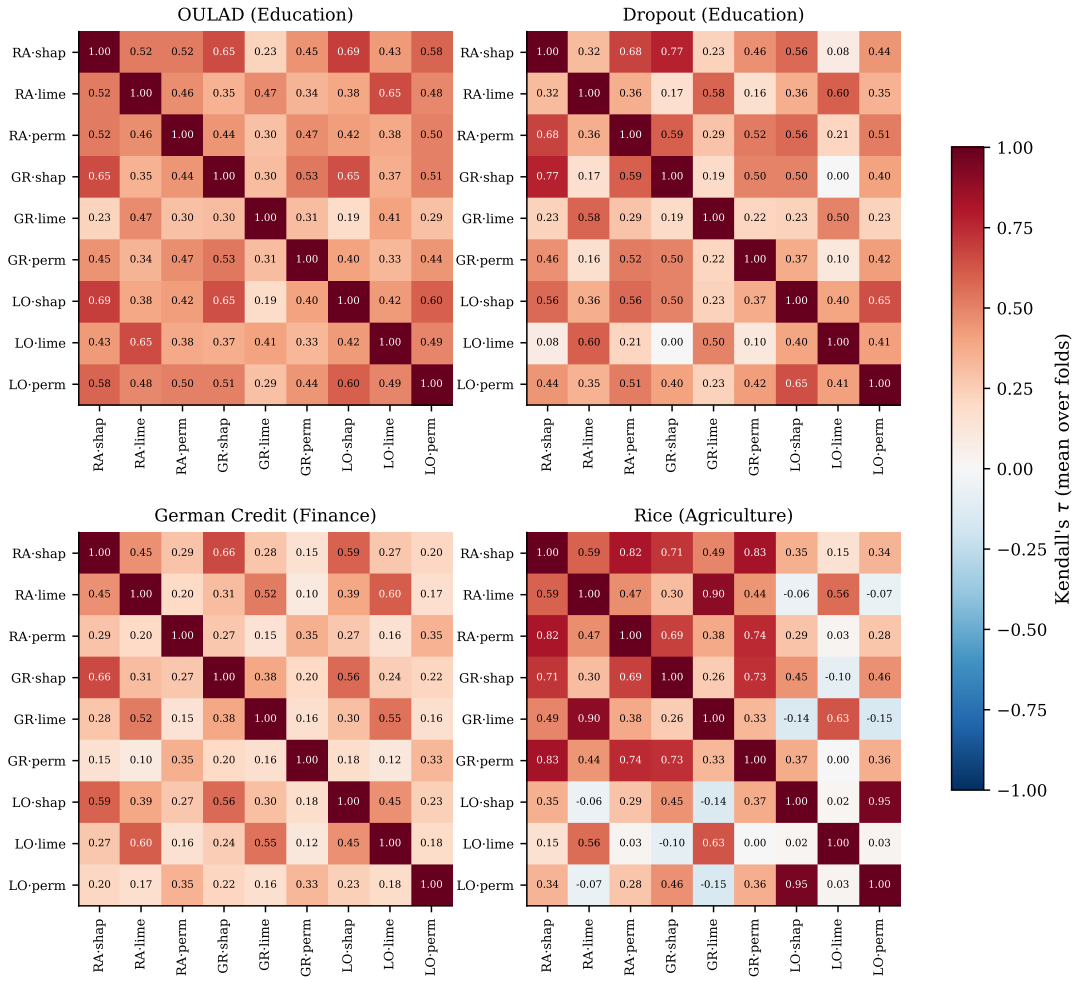


Figure 1: Pairwise Kendall's τ concordance among the nine (model, method) combinations, by dataset.

10 do not appear in LIME's top 10 at all (no connecting line), and the ordering among the features the two methods do share still differs. This is the practical face of the $\tau \approx 0.28$–0.47 figure above: a feature list handed to a stakeholder would visibly change depending on which method produced it.

5 Discussion

The consistency of this pattern across four unrelated domains (two educational datasets with different institutional contexts, a financial credit dataset, and an agricultural morphology dataset) argues against explaining it away as a quirk of one problem. A conclusion of the form "feature X is the most important driver of this model's predictions," when it rests on a single attribution method, is method-contingent: a different, equally standard method applied to the identical trained model shows top-10-restricted concordance between $\tau = 0.055$ (German Credit) and $\tau = 0.244$ (Dropout) for three of the four datasets (Results), meaning the specific features flagged as most important frequently differ between methods.

This matters most in exactly the settings where attribution methods are invoked as justification (credit decisions, admissions or dropout-risk flagging, medical triage) because the audience for such an explanation (a regulator, an affected individual, an auditor) has no way to know that the reported ranking is method-contingent unless the analyst says so. Our recommendation, consistent with what the agreement numbers support, is to report at least two attribution methods and their concordance (a single τ value is cheap to compute) rather than a single ranking presented as though it were the model's ground truth.

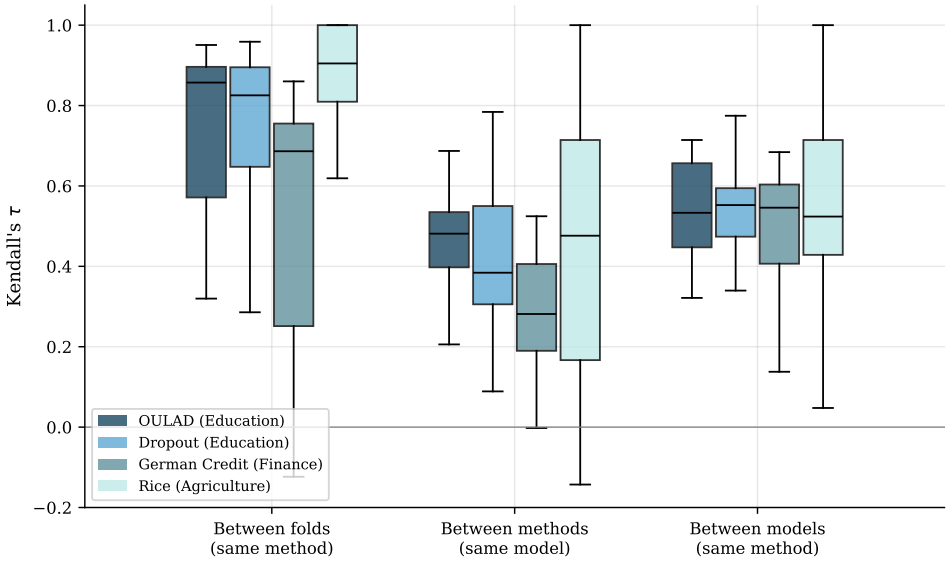


Figure 2: Kendall's τ by comparison axis and dataset.

Table 4: Two-way ANOVA of Kendall's τ on dataset and comparison axis, raw pairwise specification ($n = 2{,}340$). Partial $\eta^2 = SS_{\text{effect}}/(SS_{\text{effect}} + SS_{\text{residual}})$.

Source	Sum of squares	df	F	p	Partial η^2
Dataset	24.06	3	222.63	5.7×10^{-127}	0.223
Axis	37.98	2	527.12	1.5×10^{-189}	0.312
Dataset × Axis	4.10	6	18.95	1.2×10^{-21}	0.047
Residual	83.88	2,328			

The comparatively higher between-model agreement points to a specific mechanism: it suggests that a chosen explanation method captures something at least partially about the *data*, namely which features are genuinely informative for the task, that persists somewhat across model families, whereas the low between-method agreement suggests each method also imposes its own distinct lens (SHAP's game-theoretic allocation, LIME's local linear surrogate, permutation's direct ablation) that shapes the ranking as much as the underlying data does.

Our quantitative estimate of the disagreement's magnitude is consistent with, and adds statistical weight to, Krishna et al.'s [5] practitioner-reported finding that data scientists routinely encounter and informally work around explanation disagreement: where their evidence is that practitioners *notice* the problem, ours is a controlled measurement of *how large* it is and that it does not shrink as the application domain changes.

6 Limitations

This study is limited to tabular data; image and text models, where SHAP and LIME operate over different units (pixels/superpixels, tokens) rather than a shared feature space, were out of scope. We compare three widely used methods out of a substantially larger family (e.g., Integrated Gradients, counterfactual explanations, anchors were not included). LIME's aggregation from local, per-instance explanations to a single global ranking is itself a design choice with more than one reasonable implementation; we used one documented, defensible approach, but did not exhaustively test alternative aggregation schemes. Furthermore, LIME discretizes continuous features into quantiles by default to stabilize linear surrogate fitting, whereas SHAP and permutation importance evaluate features in continuous space; while this follows standard applied practice, part of the between-method divergence reflects this structural difference in perturbation geometry.

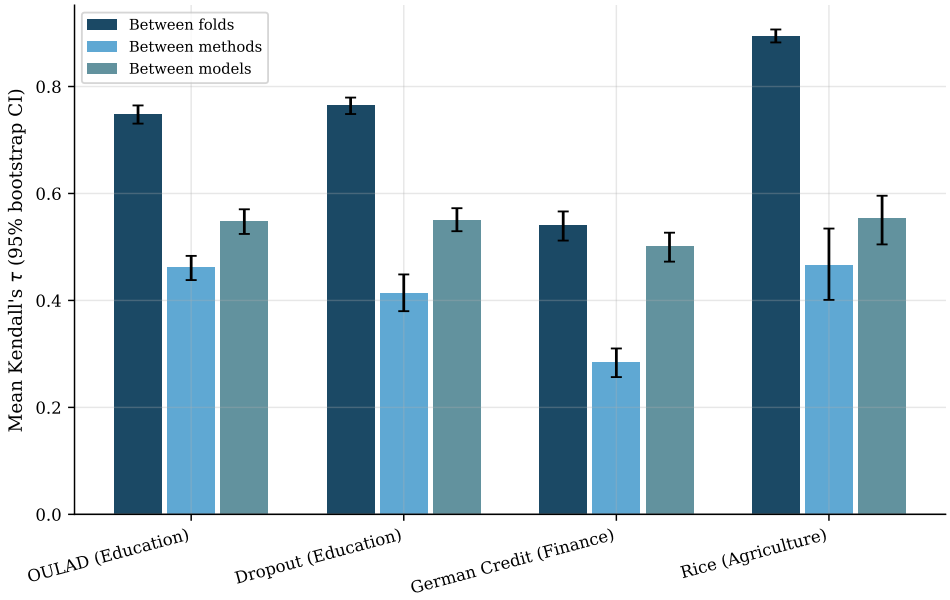


Figure 3: Kendall's τ by dataset: does the agreement pattern depend on the application domain?

In multiclass settings (OULAD and Dropout), SHAP aggregates attribution across all candidate classes, whereas LIME conditions on the predicted class; however, on strictly binary tasks (German Credit and Rice), where this outcome asymmetry does not exist, inter-method agreement remains equally suppressed ($\tau \in [0.28, 0.47]$), demonstrating that the disagreement is fundamental rather than an artifact of class-aggregation asymmetry. Preprocessing (standardization and imputation) was applied uniformly; tree-based ensembles (Random Forest and XGBoost) are invariant to monotonic scaling, and identical pipeline execution across all folds ensures that preprocessing variations cannot account for the observed inter-method disagreement (mean intra-method fold stability reached $\tau = 0.89$ on Rice).

The between-model axis additionally confounds model family with explainer algorithm, since SHAP uses `TreeExplainer` for the tree-based models and `LinearExplainer` for logistic regression; disagreement attributed to "model" on that axis may partly reflect this estimator change. We checked this directly: if the explainer swap were inflating SHAP's between-model disagreement, SHAP should show the *lowest* between-model τ of the three methods, since it is the only one whose estimator changes across models. It shows the highest instead (mean between-model τ: SHAP = 0.595, LIME = 0.580, permutation importance = 0.439, pooled across datasets and folds), which runs opposite to what the confound would predict and is more consistent with permutation importance, not SHAP, being the outlier on this axis for reasons unrelated to the explainer-swap concern. This does not rule out every possible effect of the estimator change, but it argues against it being a major driver of the reported between-model figures.

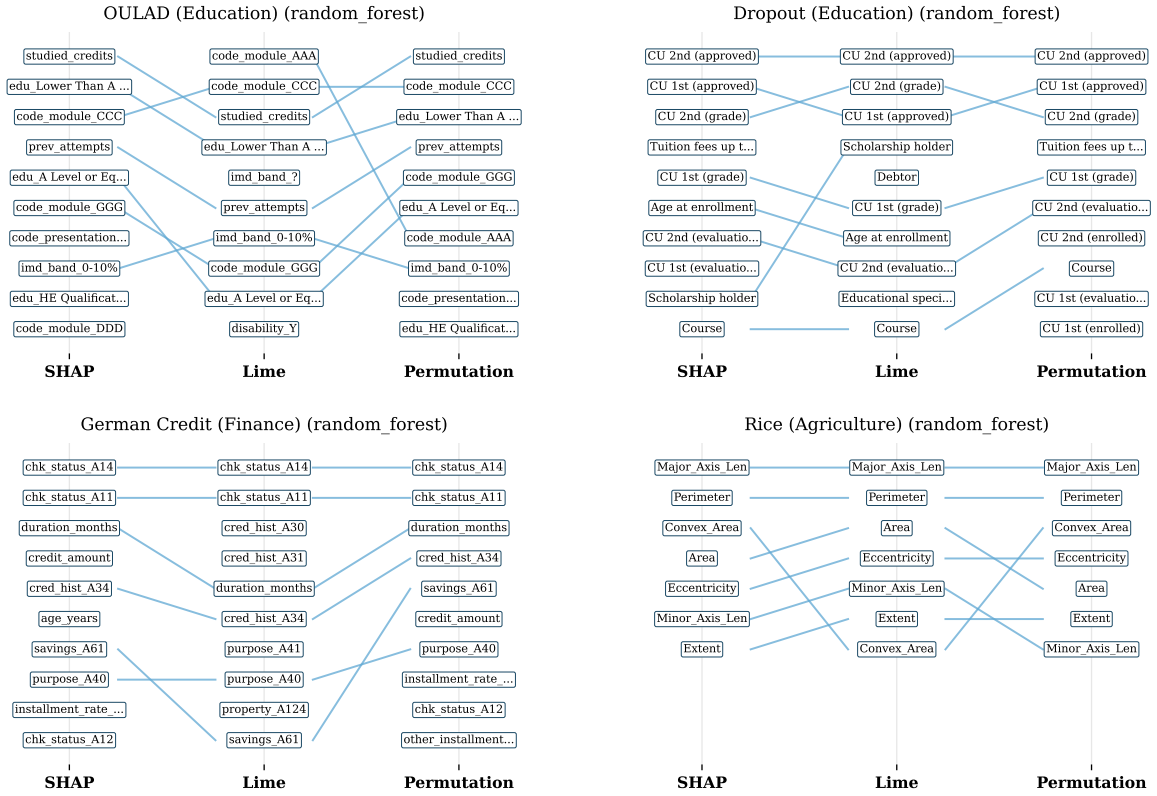


Figure 4: Top-10 features by SHAP, LIME, and permutation importance (random forest), all four datasets; lines connect a feature that appears in the top 10 of both adjacent methods.

7 Conclusion

Across four datasets from unrelated domains, three model families, and ten cross-validation folds, feature-attribution methods that are routinely treated as interchangeable diagnostic tools produce rankings with mean $\tau \in [0.28, 0.47]$ against each other (64%–74% concordant feature pairs), consistently lower than their agreement with themselves across resampling folds ($\tau \in [0.54, 0.89]$, 77%–95% concordant pairs), and lower still when the comparison is restricted to the top-10 features most likely to be reported to a stakeholder. Choice of explanation method has a measurable effect on the resulting ranking, of a magnitude that a repeated-seed control shows is not attributable to estimator noise. We recommend reporting concordance between at least two attribution methods, rather than the output of a single one, for explainability claims used to justify decisions.

Data Availability

The preprocessed datasets, trained models, attribution rankings, and analysis code are permanently archived and openly accessible in Zenodo at https://doi.org/10.5281/zenodo.22907629 and GitHub at https://github.com/JuanAvilesP3/informatica-concordancia-xai.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Author Contributions

Italo Javier Tenempaguay-Granizo: Conceptualization, Methodology, Formal Analysis, Writing - Original Draft. **Juan Pablo Aviles-Esparza:** Software, Validation, Investigation, Writing - Review & Editing. **Isaac David Torres-Paredes:** Supervision, Project Administration, Resources, Writing - Review & Editing.

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References

[1] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 4765–4774, 2017. [Online], Available: https://doi.org/10.48550/arXiv.1705.07874.

[2] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should I trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016. [Online], Available: https://doi.org/10.1145/2939672.2939778.

[3] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *Journal of Machine Learning Research*, vol. 20, no. 177, pp. 1–81, 2019. [Online], Available: https://doi.org/10.48550/arXiv.1801.01489.

[4] Z. C. Lipton, "The mythos of model interpretability," *Queue*, vol. 16, no. 3, pp. 31–57, 2018. https://doi.org/10.1145/3236386.3241340.

[5] S. Krishna, T. Han, A. Gu, J. Pombra, S. Jabbari, S. Wu, and H. Lakkaraju, "The disagreement problem in explainable machine learning: A practitioner's perspective," *Transactions on Machine Learning Research*, 2024. [Online], Available: https://doi.org/10.48550/arXiv.2202.01602.

[6] D. Alvarez-Melis and T. S. Jaakkola, "On the robustness of interpretability methods." ICML 2018 Workshop on Human Interpretability in Machine Learning (WHI), 2018. arXiv:1806.08049, https://arxiv.org/abs/1806.08049.

[7] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning." arXiv preprint, 2017. arXiv:1702.08608, https://arxiv.org/abs/1702.08608.

[8] D. Li, "Mining key influencing factors of explanatory AI in teaching digital decision support system," *Informatica*, vol. 49, no. 26, pp. 1–10, 2025. [Online], Available: https://doi.org/10.31449/inf.v49i26.10890.

[9] W. Yao and J. Zhu, "Optimizing public hospital budgets using ensemble machine learning and SHAP analysis for interpretable cost prediction," *Informatica*, vol. 49, no. 22, pp. 1–11, 2025. [Online], Available: https://doi.org/10.31449/inf.v49i22.7981.

[10] D. Vlahek, "An efficient iterative algorithm to explainable feature learning," *Informatica*, vol. 48, no. 2, pp. 1–12, 2024. [Online], Available: https://doi.org/10.31449/inf.v48i2.6105.

[11] C. Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. Independently published, 3 ed., 2025. Available: https://christophm.github.io/interpretable-ml-book/.

[12] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES '20)*, pp. 180–186, 2020. [Online], Available: https://doi.org/10.1145/3375627.3375830.

[13] P.-J. Kindermans, S. Hooker, J. Adebayo, M. Alber, K. T. Schütt, S. Dähne, D. Erhan, and B. Kim, "The (un)reliability of saliency methods," in *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, vol. 11700 of *Lecture Notes in Computer Science*, pp. 267–280, Springer, 2019. [Online], Available: https://doi.org/10.1007/978-3-030-28954-6_14.

[14] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, and C. Seifert, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable AI," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 295:1–295:42, 2023. [Online], Available: https://doi.org/10.1145/3583558.

[15] G. Visani, E. Bagli, and F. Chesani, "Statistical stability indices for LIME: Obtaining reliable explanations for ML models," *Journal of the Operational Research Society*, vol. 73, no. 1, pp. 91–101, 2022. [Online], Available: https://doi.org/10.1080/01605682.2020.1865846.

[16] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, pp. 206–215, 2019. [Online], Available: https://doi.org/10.1038/s42256-019-0048-x.

[17] T. Speith, "A review of taxonomies of explainable artificial intelligence (XAI) methods," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pp. 2239–2250, 2022. [Online], Available: https://doi.org/10.1145/3531146.3534639.

[18] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. [Online], Available: https://doi.org/10.1023/A:1010933404324.

[19] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016. [Online], Available: https://doi.org/10.1145/2939672.2939785.